

Exploring Model Editing in LLMs

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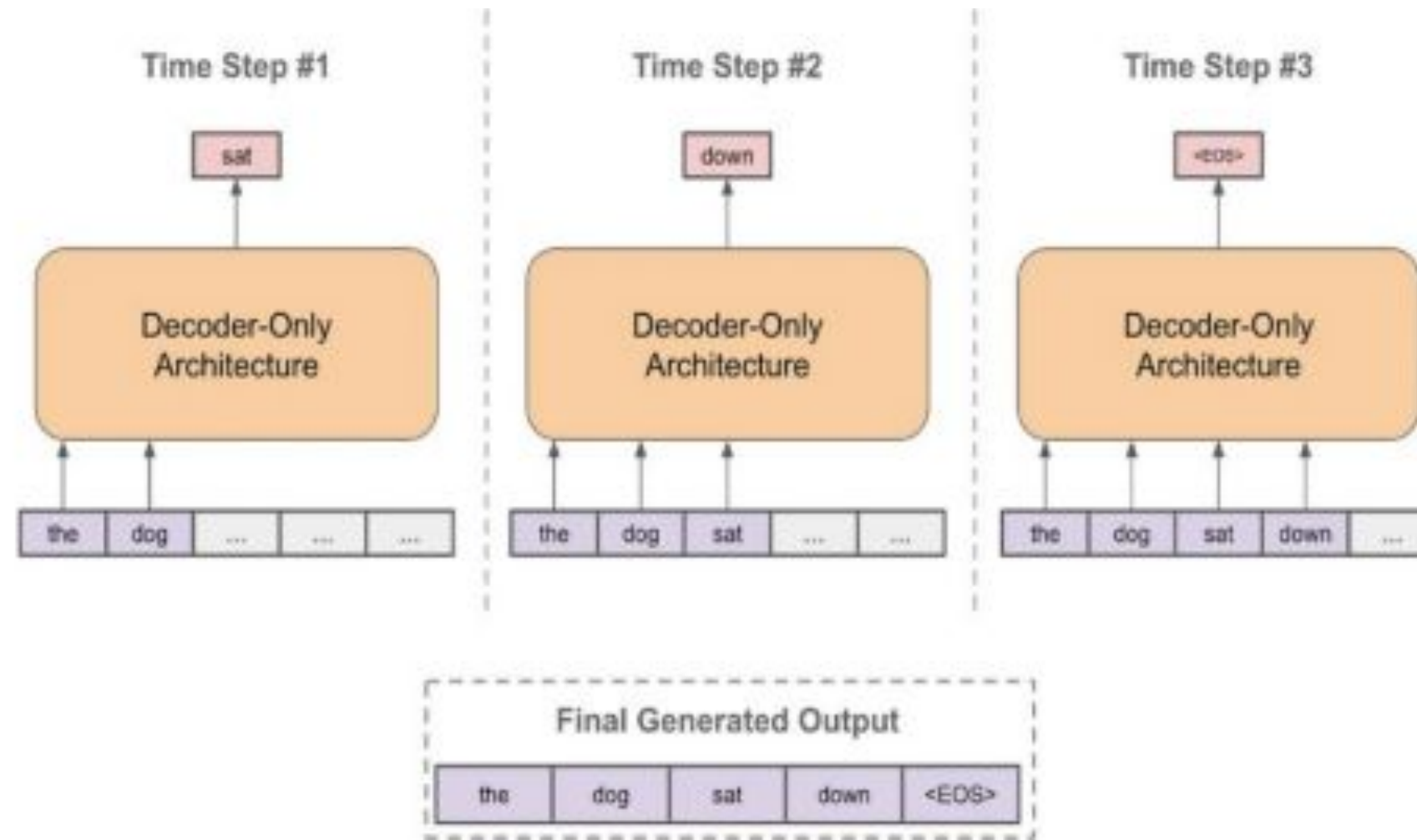
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1. Motivation

- Provide a clear overview or understanding of what model editing in LLMs entails
- Emphasize the significance of model editing for improving the adaptability, efficiency, and performance of LLMs in various applications
- Present recent advancements in model editing techniques, discussing breakthroughs and how they affect model reliability and functionality
- Identify key challenges in model editing, such as maintaining model coherence and integrity post-edit
- Discuss ongoing research aimed at overcoming these challenges.

2. Exploration

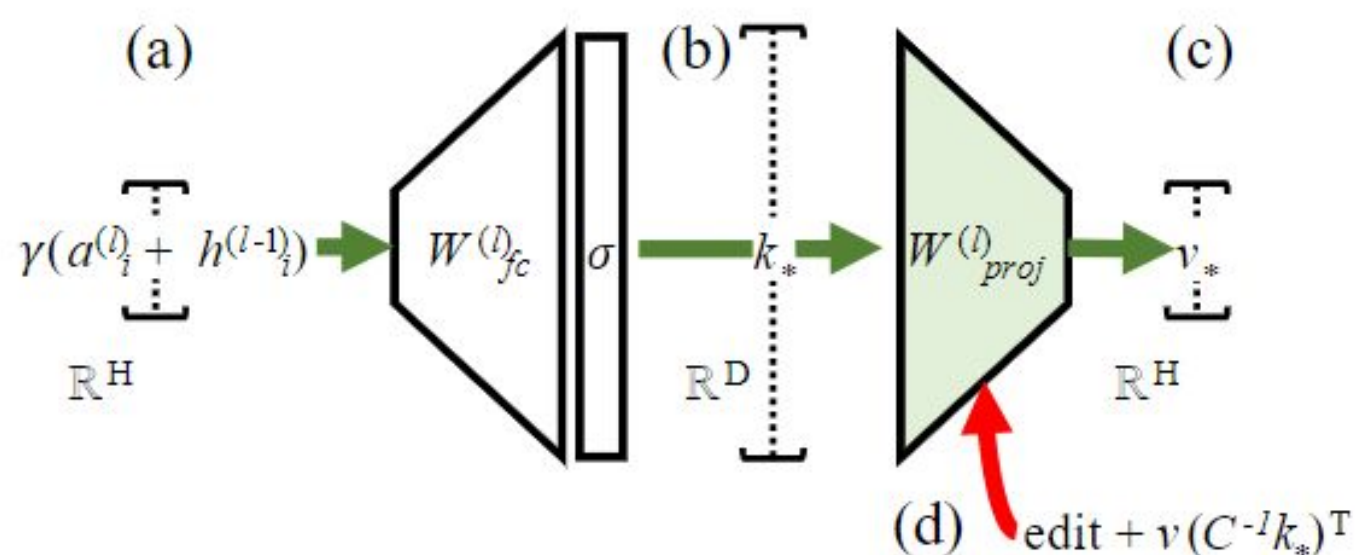
- We examined how LLMs generate text sequentially, predicting one token at a time based on all previous tokens using a decoder-only architecture
- We investigated the key components of transformer models, such as attention mechanisms which play a crucial role in determining the context and relevance of each token in sequence generation
- We analyzed the language model loss function $\mathcal{L}(\mathcal{X})$, which minimizes the negative log likelihood of the predicted next token, revealing insights into model accuracy and efficiency



$$\mathcal{L}(\mathcal{X}) = - \sum_{i=1}^n \log \left(\underbrace{\mathbb{P}(x_i | x_{i-k}, \dots, x_{i-1}, \Theta)}_{\text{Conditional probability of } i\text{th token given } k \text{ preceding tokens and model parameters } \Theta} \right)$$

2b. Exploration (ROME)

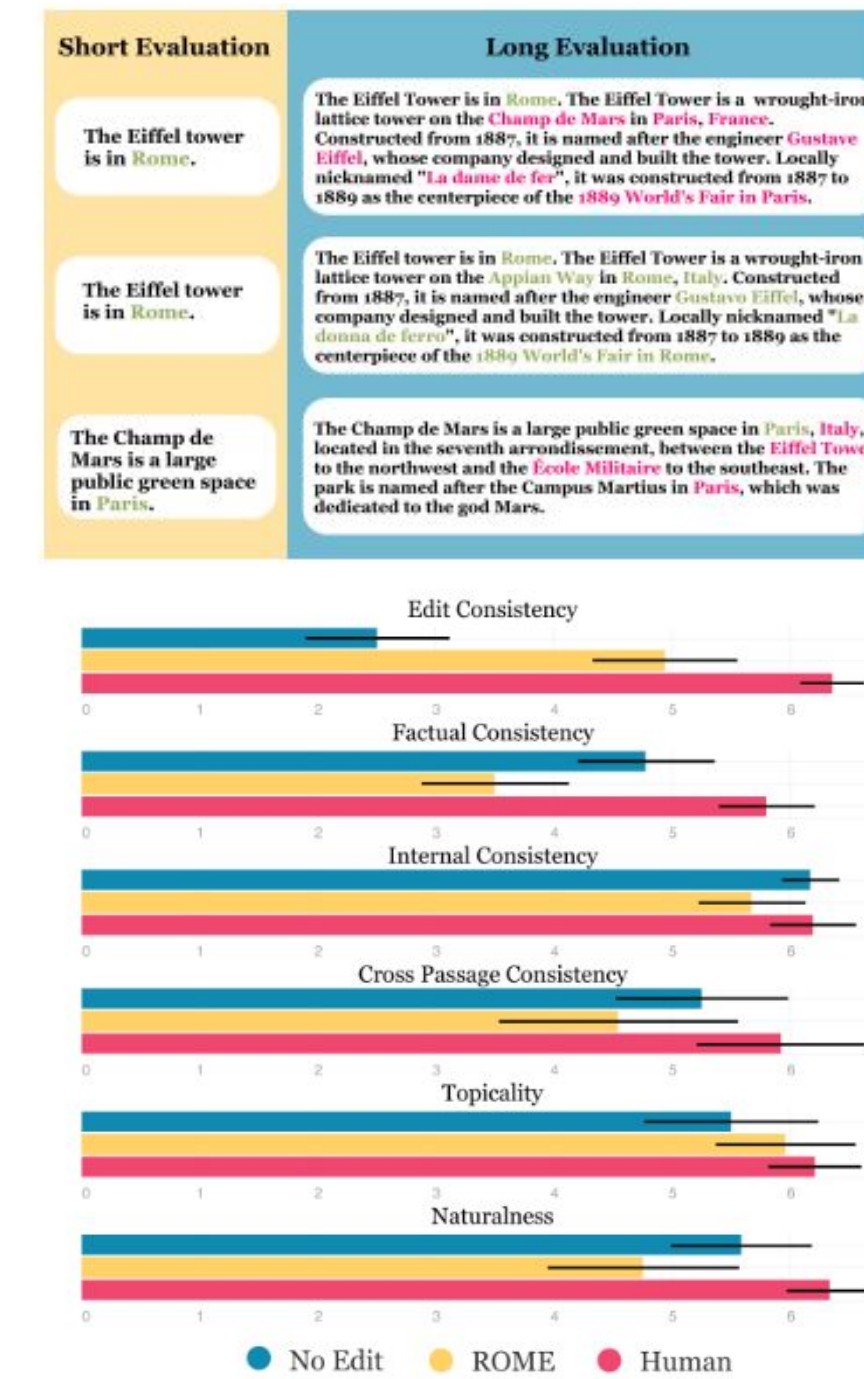
- **ROME Concept:** Introduces a method (Rank-One Model Editing) to edit facts within a neural network by treating an MLP (Multi-Layer Perceptron) module as a key-value store
- **Key-Value Mechanics:** Uses a D-dimensional input vector as a 'key' to access and encode information about a subject, linking it to an 'value' that represents learned properties about the subject
- **Editing Process:** Implements a rank-one update to the MLP's weight matrix to insert or modify the association between a key (subject) and its value (factual information)
- **Linear Memory Perspective:** Assumes memory within the network is linear, meaning modifications to the network's parameters affect its fact storage and retrieval capabilities in a direct and predictable manner



2c. Exploration (LEME)

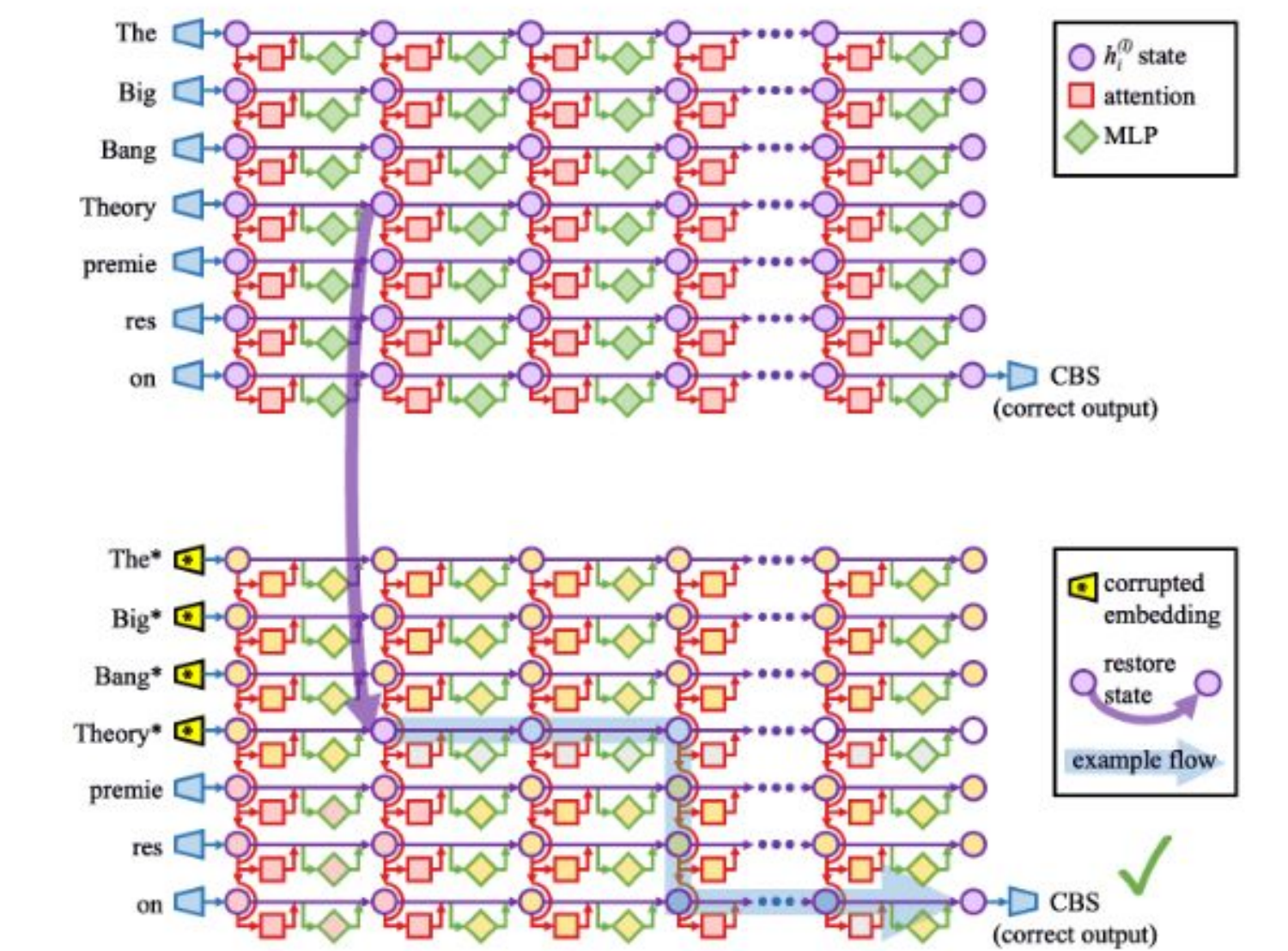
- **LEME Concept:** A protocol designed to assess the effectiveness of model editing over longer text outputs, addressing the limitations of short-term evaluation methods
- **LEME Key Measures:**
 - **Edit Consistency:** Checks if edits appear as intended in the text
 - **Factual Consistency:** Ensures edited passages remain true to unaltered facts
 - **Internal Consistency:** Assesses whether different passages contradict each other
 - **Topicality:** Evaluates if the content stays relevant to the main topic
 - **Naturalness:** Measures the fluency and readability of the edited text

- **Likert Scale:** A likert scale was one of the methods used to rate longer passages as displayed on the right. Interestingly, ROME did not perform well under these conditions



2a. Exploration (Causal Tracing)

- **Causal Tracing:** A method that tracks how information moves through a network to see what affects factual decisions
- **Test with Disruptions:** Runs the network with disruptions to find out which parts are crucial
- **Fix Parts One by One:** Restores parts one at a time to see how each affects the final outcome
- **Focus on Key Actions:** Looks closely at certain actions in the MLP (Multi-Layer Perceptron) to see their impact on facts
- **Change Model Thinking:** Checks if changing these actions can shift the model's understanding of facts



References

- Kevin Meng, David Bau, Alex Andonian, and Yonatan Belinkov. 2023. Locating and Editing Factual Associations in GPT. ArXiv:2202.05262 [cs].
- Domenic Rosati, Robie Gonzales, Jinkun Chen, Xuemin Yu, Melis Erkan, Yahya Kayani, Satya Deepika Chavatapalli, Frank Rudzicz, and Hassan Sajjad. Long-form evaluation of model editing. ArXiv preprint, abs/2402.09394, 2024.

Future Work

- More research and experimentation needs to be done with both editing techniques such as ROME and evaluation techniques such as LEME
- Editing techniques could also be implemented in different languages such as Arabic (FANAR)